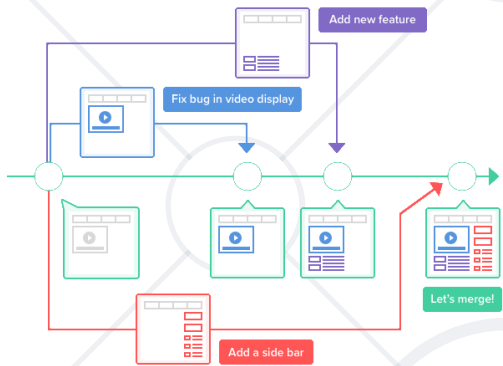


Development Workflow

Issues, Workflow, Branches, Environments



SoftUni Team
Technical Trainers



SoftUni



Software University

<https://softuni.bg>

Table of Contents

1. Issues, Branches and Environments
2. Software Development Workflow
3. Development Workflow in Action





Issues, Branches and Environments

In the Software Development Workflow

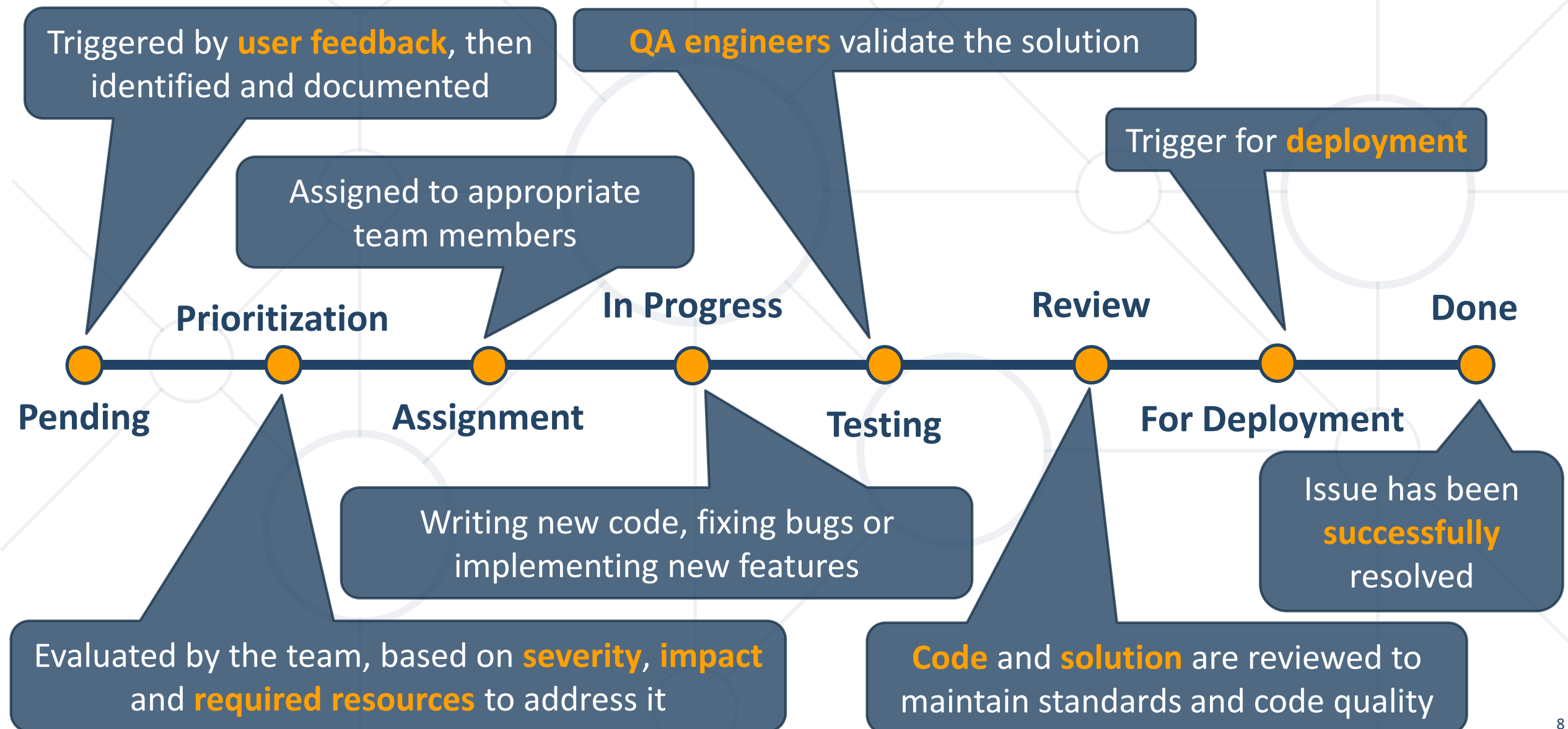
- **Set** of **processes** that a software development team uses to convert a user story into a working software
- Includes **coding**, **build** processes and automation, **testing** and **running** the application in **different environments**
- **Optimizes** work processes
- Ensures easy navigating the complexities of a software project
 - Aims for quality, reducing delays and agile development

- Development workflow includes **various stages** for addressing issues
 - Help maintain a **structured** and **organized approach** to software development
- **Issues** are usually **visualized** on a Trello board / GitHub Project Board / Jira Board / other board
 - Issues are systematically addressed in a controlled manner
- Promotes collaboration among team member, transparency and enable efficient delivery

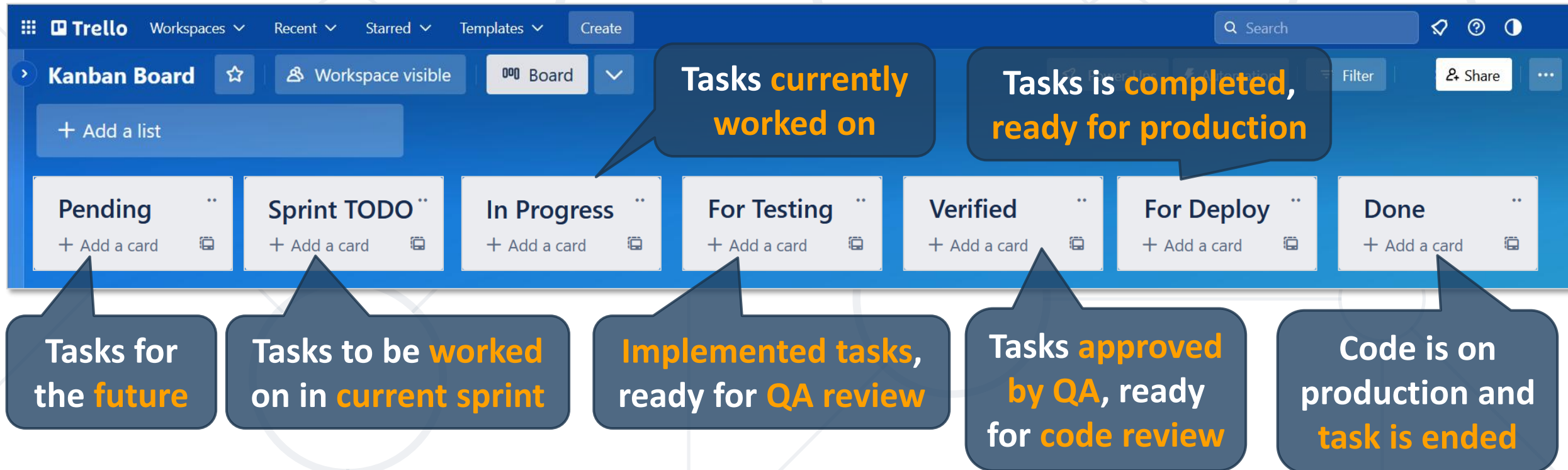
- **Pending**
 - Issues are waiting to be **addressed** and **prioritized**
- **Sprint TODO**
 - **Selected issues** to be worked on during the **current sprint**
 - **Agile** development environment
- **In Progress**
 - Issues that are **actively** being worked on
 - Developers implement the required features to address the issues

- **For Testing**
 - Ready for **QA** testing
- **Verified**
 - Testing is **complete** and ready for further actions
- **For Deploy**
 - **Verified** and approved
- **Done**
 - Successful **deployment**

Issue Lifecycle



Issues – Example



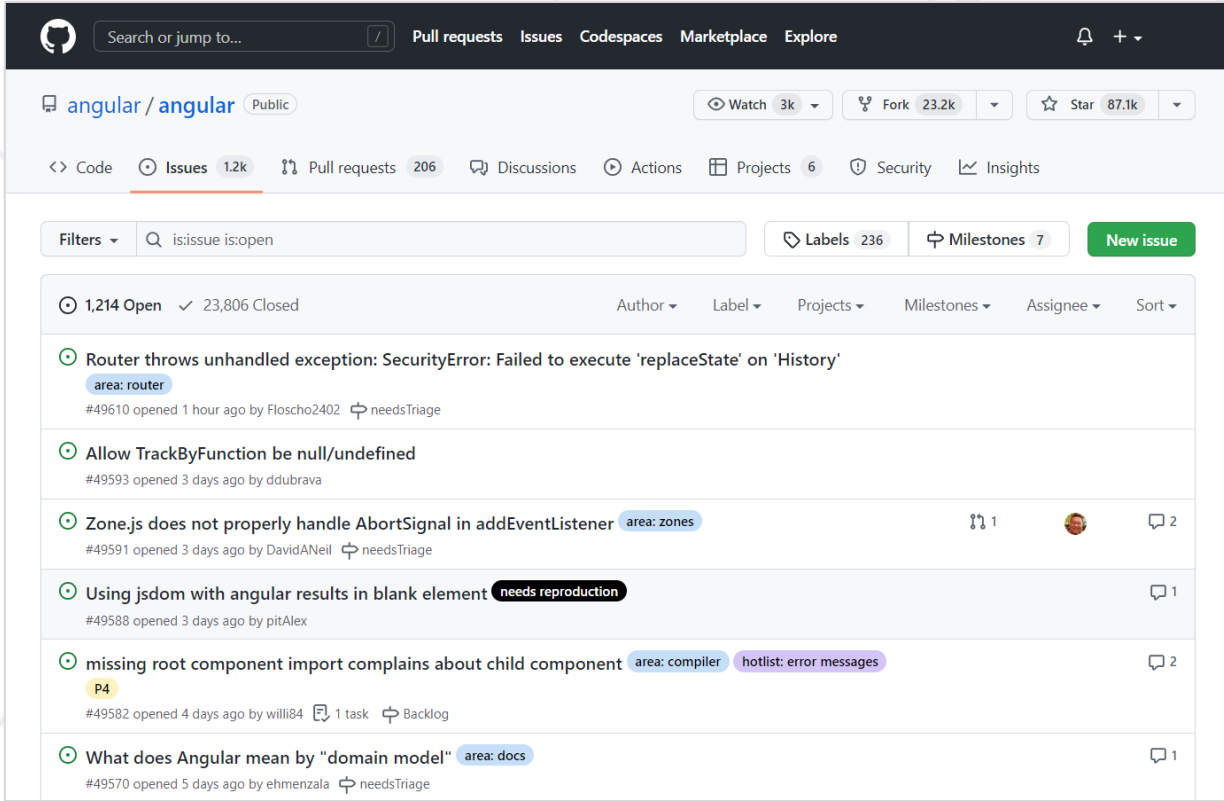
■ GitHub Issues

- Built-in feature of GitHub
- Holds **all project issues** (requirements + bugs)
- Bug reports
- Feature requests
- GitHub Issues example

■ GitHub Projects

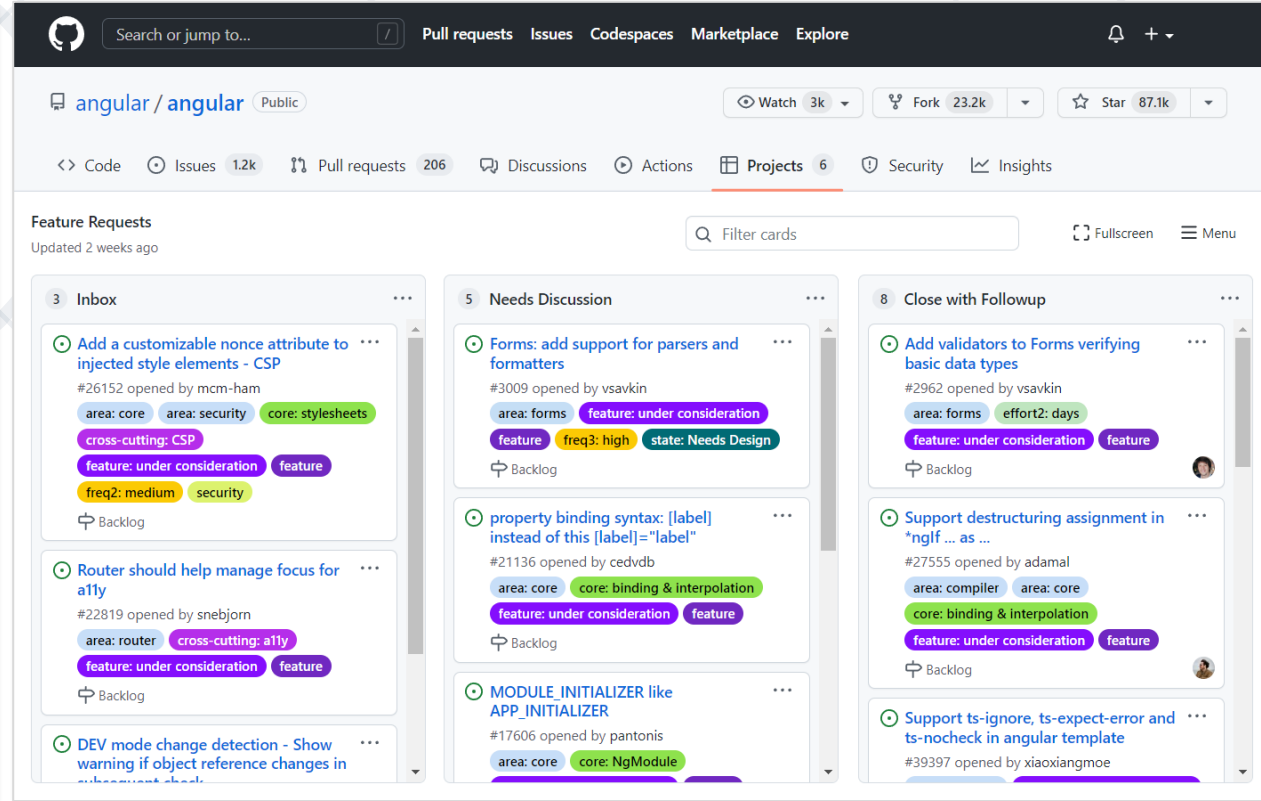
- Project management feature of GitHub
- Enables planning, organizing and tracking the progress of a software development project
- Visualizes the issues (tasks) workflow as a Kanban board
- GitHub Projects example

Issue Trackers – Example



This screenshot shows the GitHub Issues page for the angular/angular repository. The interface includes a search bar, navigation tabs for Pull requests, Issues, Codespaces, Marketplace, and Explore, and a sidebar with filters. The main content area displays a list of open issues, each with a title, area, and status. The issues are sorted by 'Open' status, showing 1,214 open and 23,806 closed issues. The issues listed include:

- Router throws unhandled exception: SecurityError: Failed to execute 'replaceState' on 'History' (area: router, #49610)
- Allow TrackByFunction be null/undefined (area: router, #49593)
- Zone.js does not properly handle AbortSignal in addEventListener (area: zones, #49591)
- Using jsdom with angular results in blank element (needs reproduction, #49588)
- missing root component import complains about child component (area: compiler, hotlist: error messages, #49582)
- What does Angular mean by "domain model" (area: docs, #49570)



This screenshot shows the GitHub Projects page for the angular/angular repository. The interface includes a search bar, navigation tabs for Pull requests, Issues, Codespaces, Marketplace, and Explore, and a sidebar with filters. The main content area displays a Kanban board with columns for different stages of the project. The columns are:

- Inbox (3 items)
- Needs Discussion (5 items)
- Close with Followup (8 items)

The items in the 'Needs Discussion' column include:

- Forms: add support for parsers and formatters (area: forms, feature: under consideration, freq3: high, state: Needs Design)
- property binding syntax: [label] instead of this [label]="label" (area: core, core: binding & interpolation, feature: under consideration, feature)
- MODULE_INITIALIZER like APP_INITIALIZER (area: core, core: NgModule)

- **Title**
- **Main body**
 - Requirements
 - Outline of the desired functionality and its use case
 - Bug description
 - Steps to reproduce the issue, expected and actual behavior

- **Labels**
 - Categorize for easy organization
- **Discussion**
 - Comments from team members
- **History**
 - Changes over time

Inside an Issue – Example

fix(core): ComponentRef.setInput only sets input when not equal to pr...
#49607

Open atscott wants to merge 1 commit into `angular:main` from `atscott:setInputTracking`

Conversation 1

Commits 1

Checks 7

Files changed 2

+33 -0

atscott commented 6 hours ago

Contributor

...vious

`ComponentRef.setInput` currently sets the input on the component regardless of the previous value the method was called with. This results in different behavior from bindings in templates, which only set inputs when the value differs in the `Object.is` check from its previous value.

BREAKING CHANGE: `ComponentRef.setInput` will only set the input on the component if it is different from the previous value (based on `Object.is` equality). If code relies on the input always being set, it should be updated to copy objects or wrap primitives in order to ensure the input value differs from the previous call to `setInput`.

atscott added `area: core` `target: major` labels 6 hours ago

atscott added this to the `v16-candidates` milestone 6 hours ago

atscott requested review from `pkozlowski-opensource` and `alxhub` 6 hours ago

angular-robot bot added the `detected: breaking change` label 6 hours ago

fix(core): ComponentRef.setInput only sets input when not equal to pr... 56f3a89

atscott force-pushed the `setInputTracking` branch from `b3bee60` to `56f3a89` 6 hours ago

ajafff reviewed 2 hours ago

View changes

Compare

packages/core/src/render3/component_ref.ts

```
... @@ -270,8 +271,17 @@ export class ComponentRef<T> extends AbstractComponentRef<T> {
270 271     const inputData = this._tNode.inputs;
271 272     let dataValue: PropertyAliasValue|undefined;
272 273     if (inputData != null && (dataValue = inputData[name])) {
274 +      this.setInputHistory ??= {};

```

ajafff 2 hours ago

consider using `Object.create(null)` and `Object.prototype.hasOwnProperty.call(this.setInputHistory, name)` to avoid issues with special names like `__proto__` or `hasOwnProperty`

Reviewers

ajafff

pkozlowski-opensource

alxhub

At least 1 approving review is required to merge this pull request.

Still in progress? Learn about draft PRs

Assignees

No one assigned

Labels

`area: core` `detected: breaking change` `target: major`

Projects

None yet

Milestone

v16-candidates

Development

Successfully merging this pull request may close these issues.

None yet

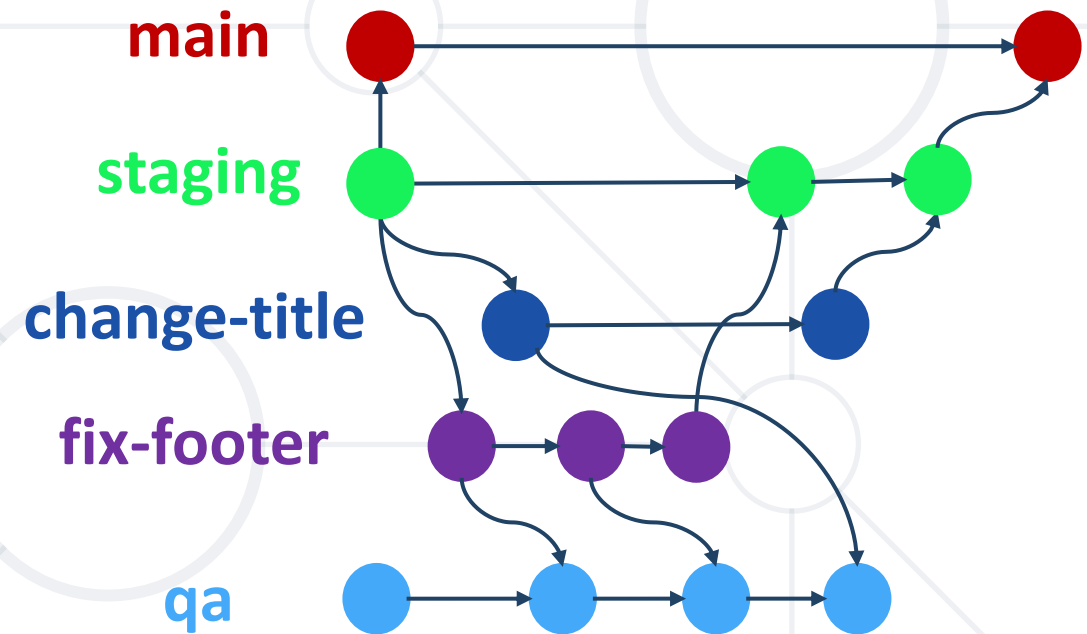
Notifications

Subscribe

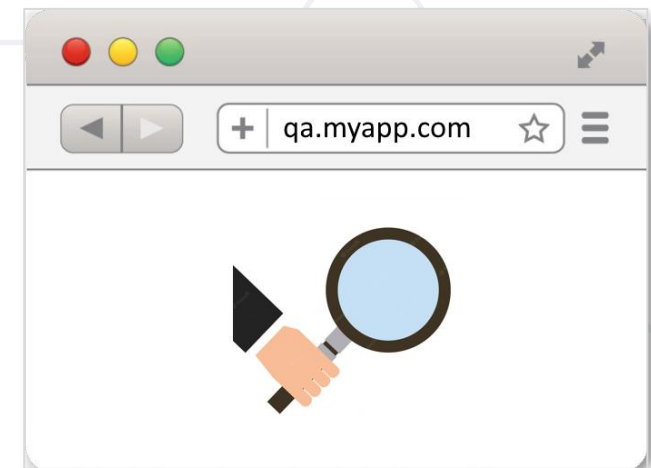
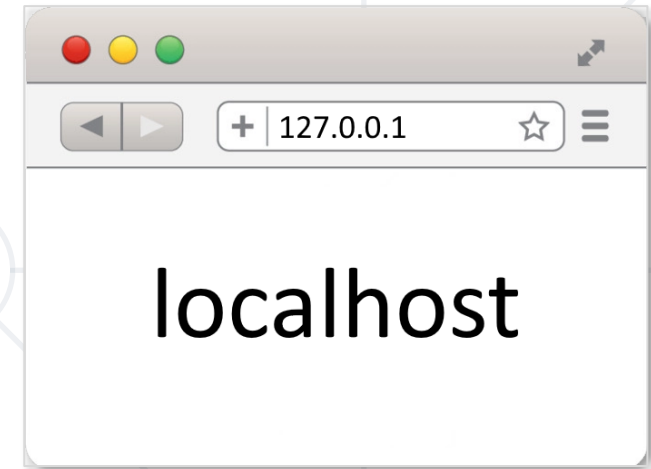
You're not receiving notifications from this thread.

2 participants

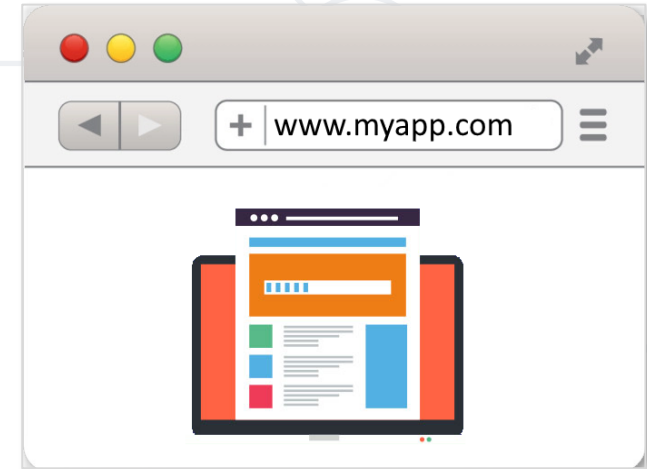
- **Feature** branch, e.g. *change-title*, *fix-footer*
 - Holds feature / bug fix under development
- **QA** branch: *qa*
 - Holds completed features, ready for testing by QAs
- **Staging** branch: *staging*
 - Holds completed and approved features, ready for production
- **Production** branch: *main*
 - Holds the production code (already deployed)



- **Local environment** (developer's laptop, localhost)
 - Where developers write code and do updates (add features, fix bugs, etc.)
- **QA environment** (testing)
 - Where QAs perform tests (automated or manual) on new or changed code



- **Staging environment** (pre-production)
 - Nearly exact replica of the production environment
 - The purpose is to test on a near-production level but in a non-production environment to verify the app will behave correctly after deployment
- **Production environment**
 - Software is running on production server; it's live for real users





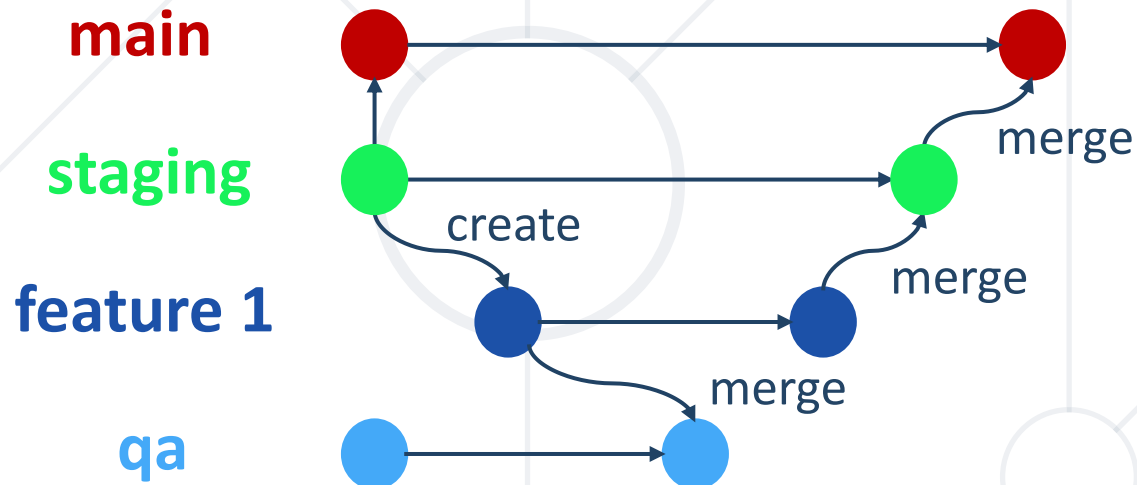
Software Development Workflow

What Happens at Each Step?

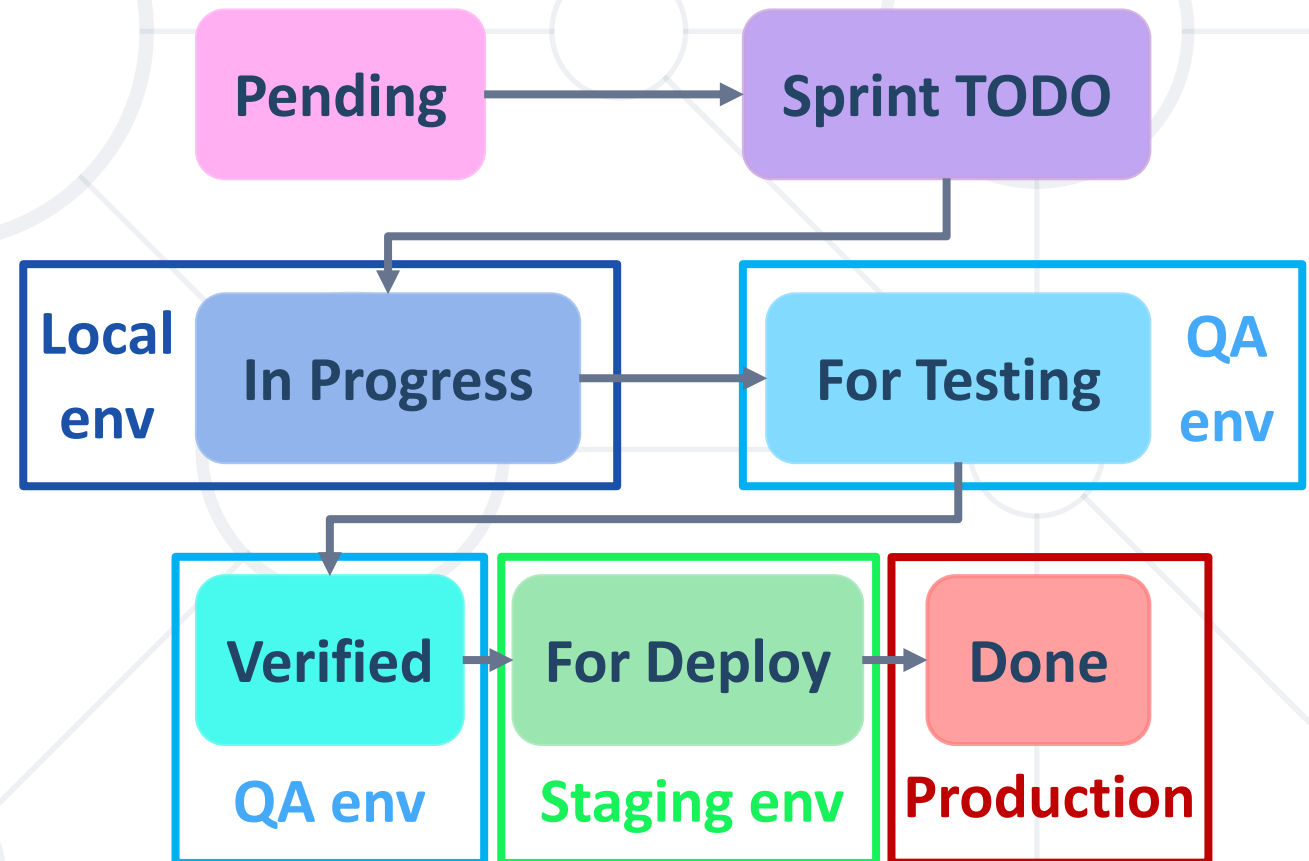
Software Development Workflow

- **Feature** branch code is available only on **local environment**
- **QA** branch code is deployed to the **QA** environment
- **Staging** branch → **staging** env.
- **Main** branch → **production** env.

Branches

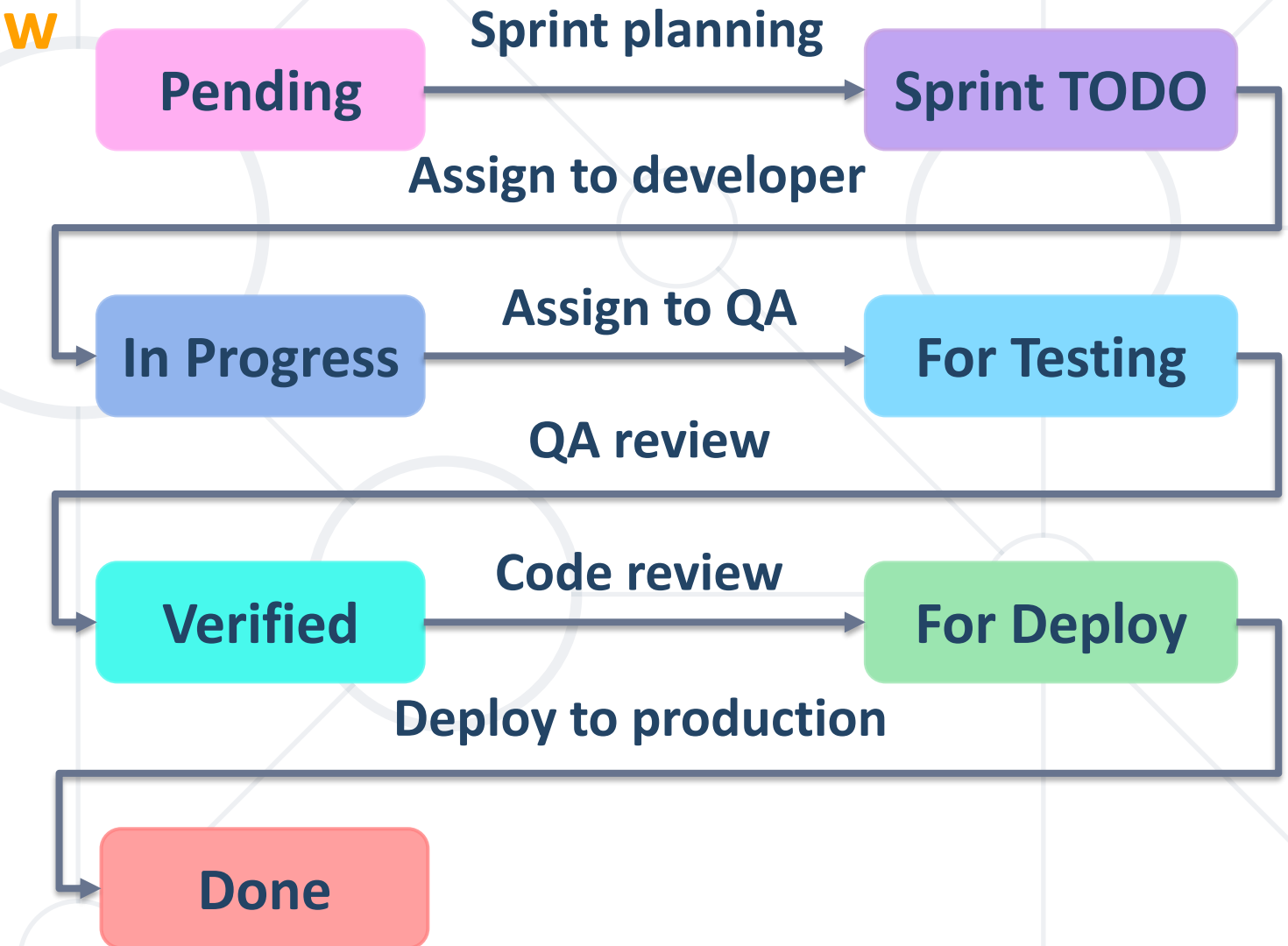


Issue Workflow



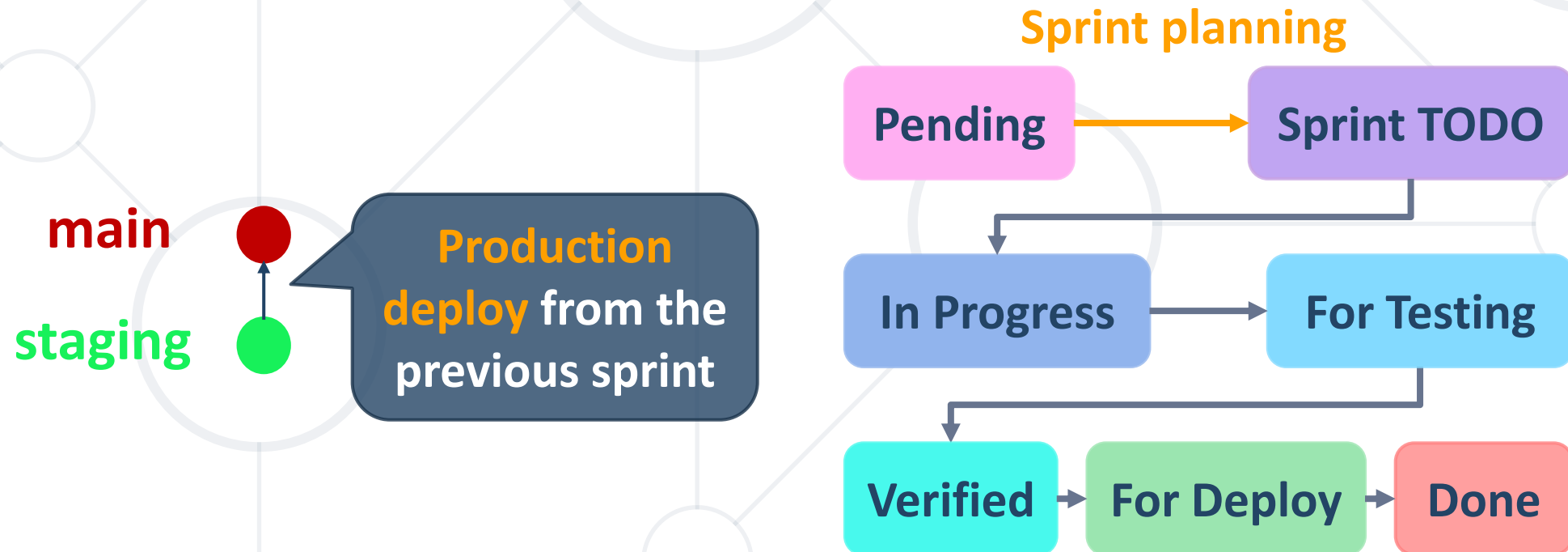
Development Workflow Steps

- The **development workflow** consists of **several steps**
- They define
 - When **issues are moved**
 - When **branches are created and merged**
 - In which **environment** dev's last changes are



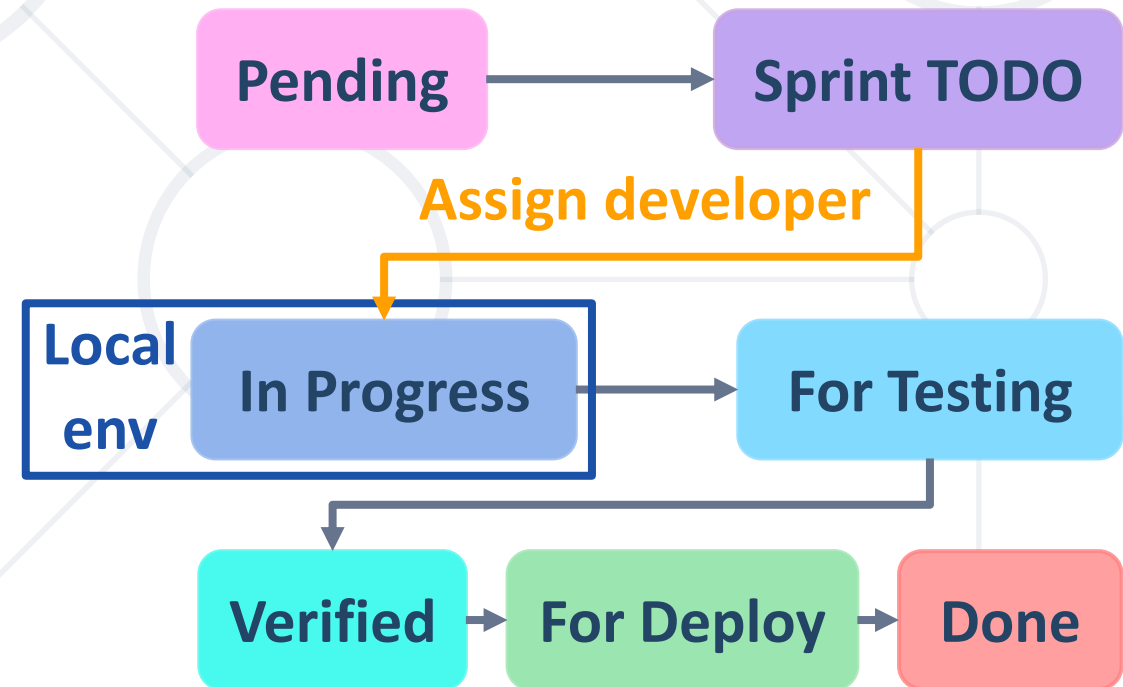
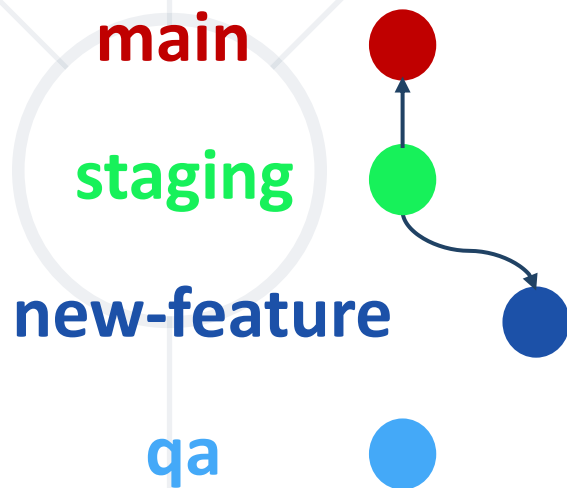
Step 1: Sprint Planning

- The most **high-priority issues** (tasks) are selected **to be implemented** during the **following sprint** (e.g., this week)
- Issues are moved from "**Pending**" to "**Sprint TODO**" column



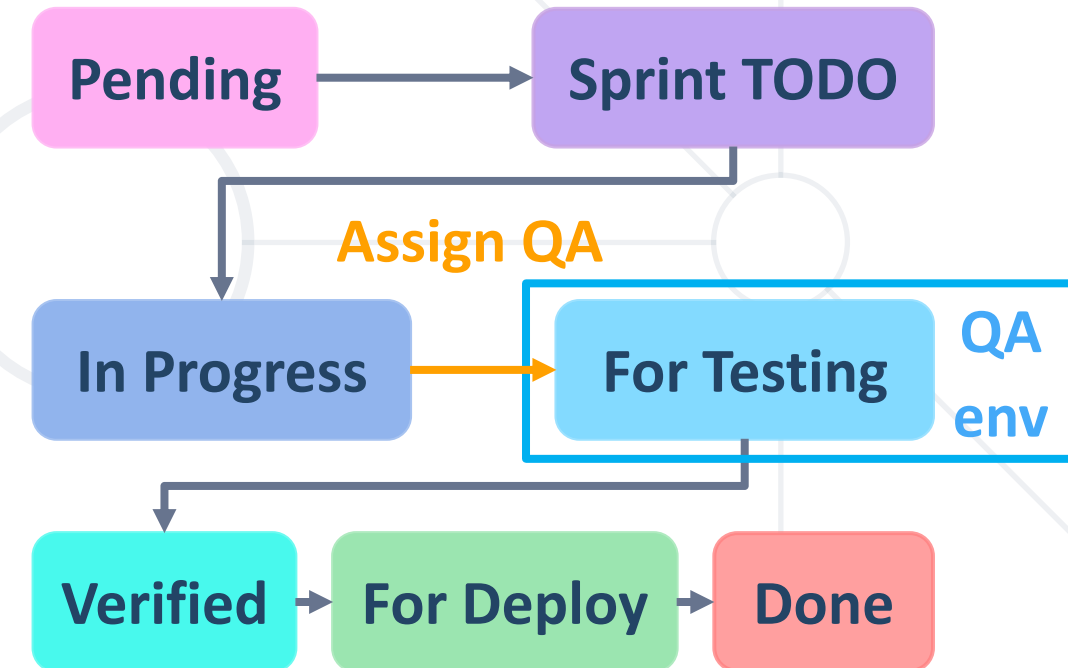
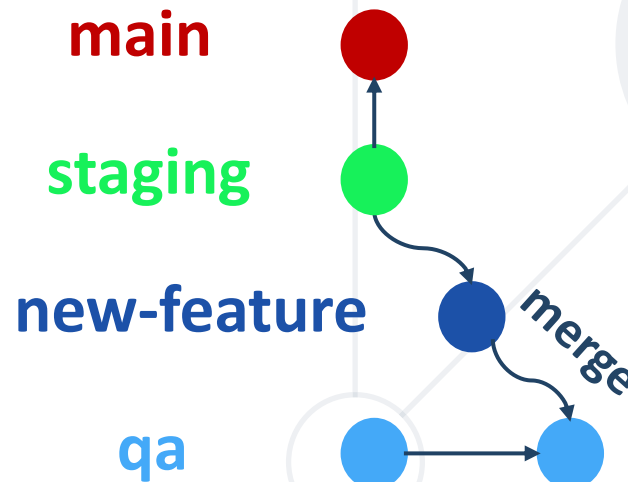
Step 2: Assign a Developer to Work on an Issue

- First, each **issue is assigned to a developer** to work on
- When developer **starts working** on it: "**Sprint TODO**" → "**In Progress**"
- Dev creates a **feature branch**
- Issue label: "**In Progress**"
- Environment: **local** (localhost)



Step 3: Assign Completed Task to QA

- Developer has **completed their task** in their **feature branch**
- Issue: "**In Progress**" → "**For Testing**" and **assigned to a QA** for testing
- Feature branch** is **merged** into the **QA branch**
- Issue label: "**For Testing**"
- Code is deployed to **QA environment**
 - Automated tests should pass
 - Often done automatically



Step 4: QA Review

- QA tests the app on the QA environment

- No bugs are found

- "For testing" → "Verified"

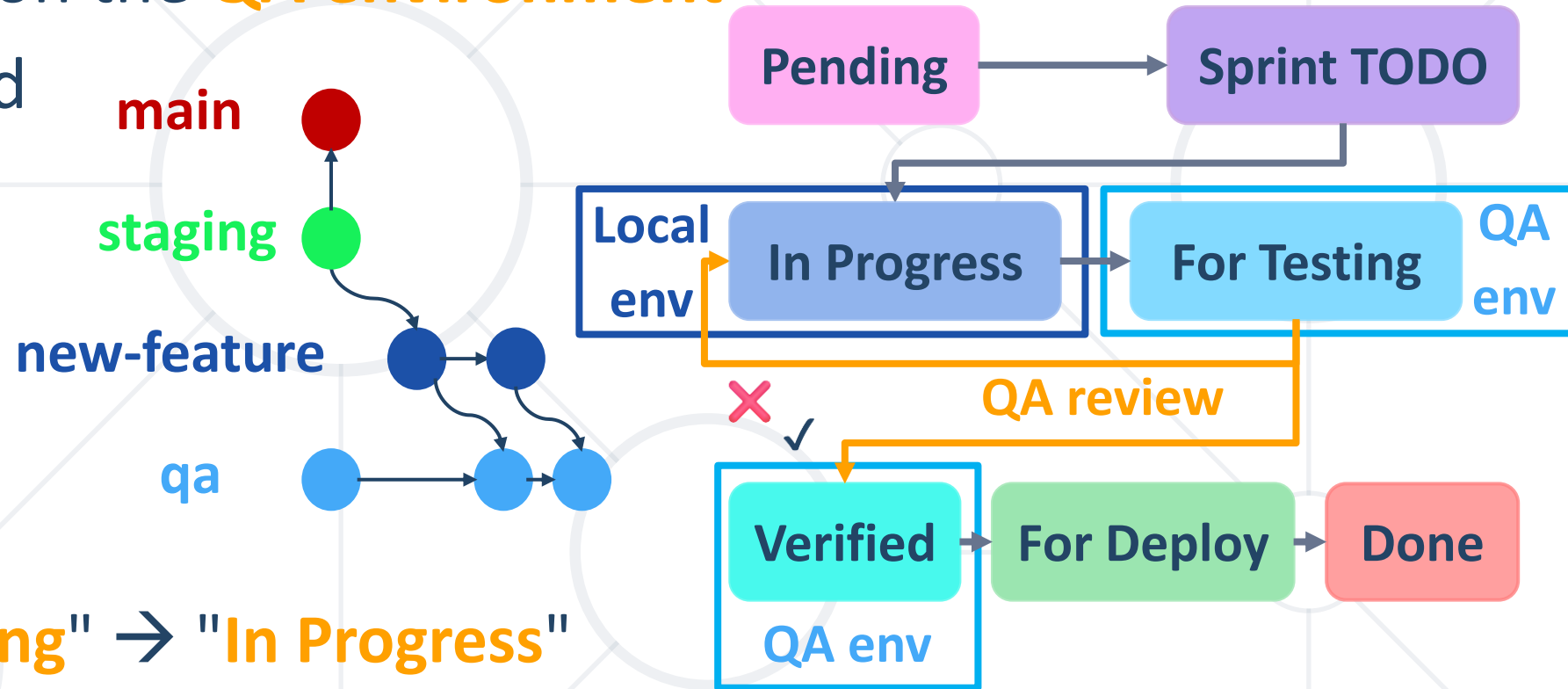
- Issue label: "QA Verified"

- If a bug is found

- Issue: "For Testing" → "In Progress"

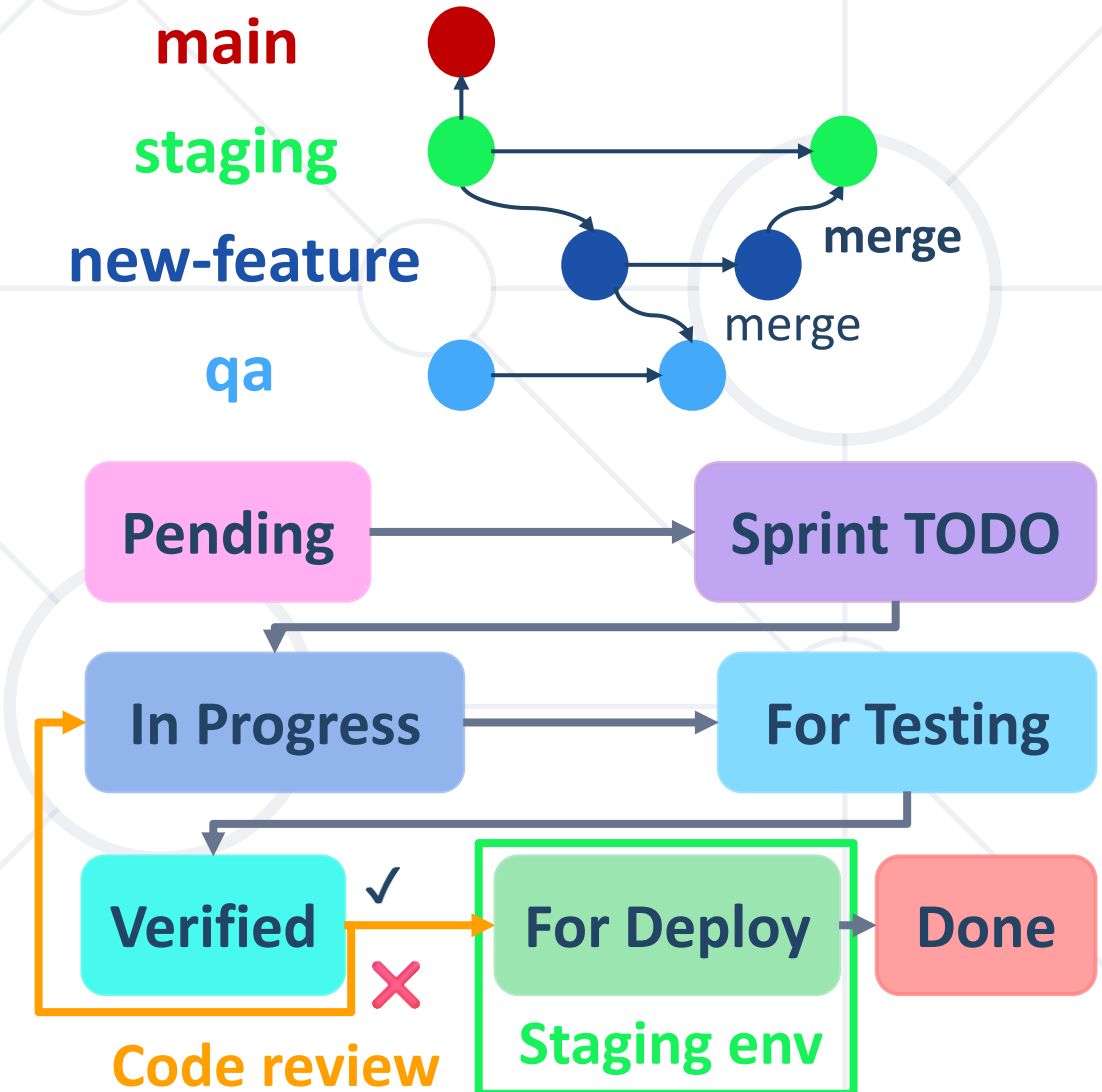
- Issue label: "QA Failed"

- Developer should make changes in their feature branch, then create a new merge into the QA branch, so that tests are performed again



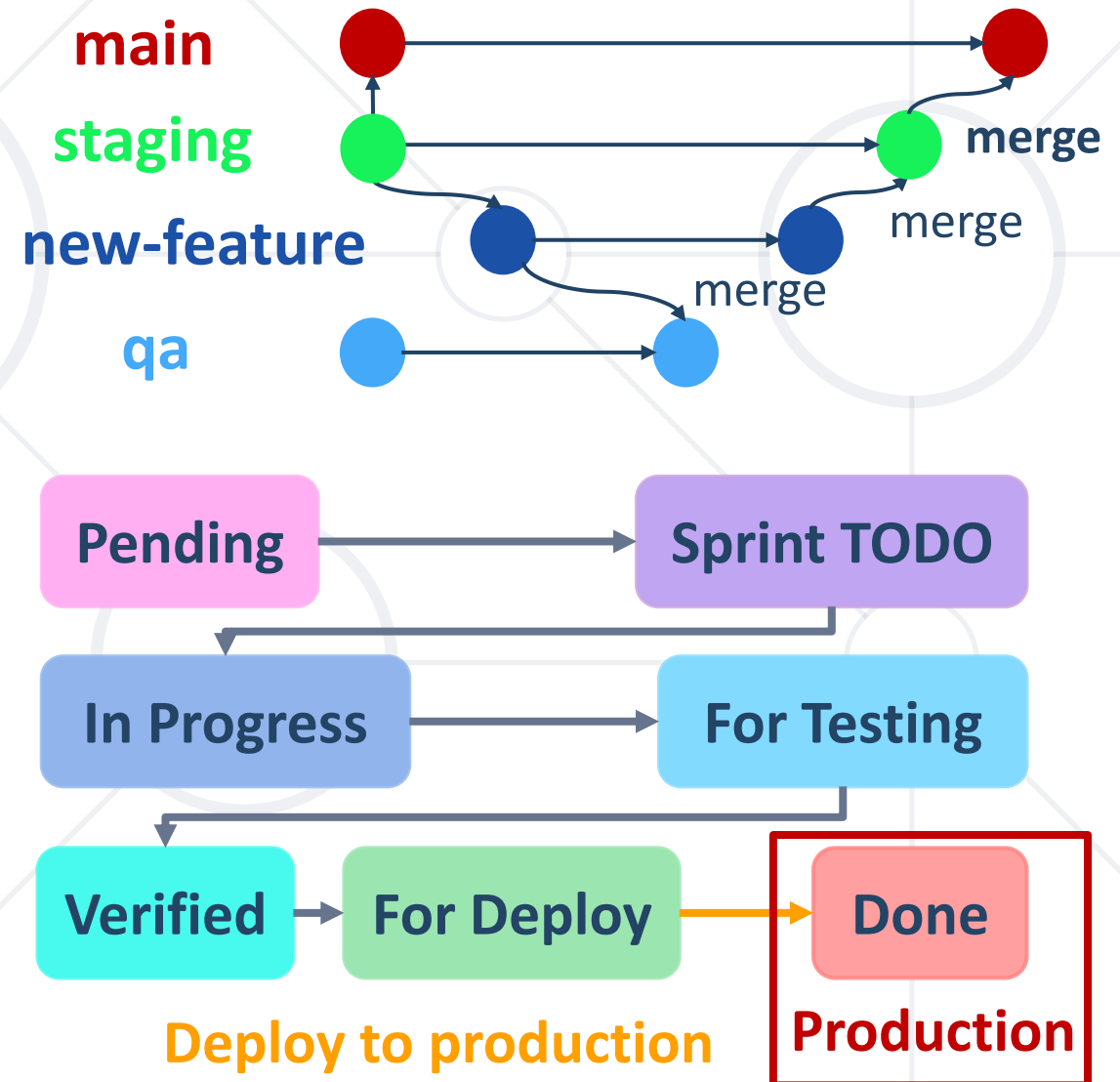
Step 5: Code Review

- Dev creates a **pull request** to **merge from the feature to staging branch**
- Other devs conduct a **code review**
- Code review **failed**
 - Issue is moved to "**In Progress**"
 - Label: "**Code Review Failed**"
- Code review **passed**
 - Label and column: "**For Deploy**"
 - Code is merged to the **staging branch** and deployed to **staging env**.
 - **Close** the issue & delete the branch



Step 6: Deploy to Production

- **Deployment to production** happens at the end of the sprint (e.g., once per week)
- Person from the team creates a **pull request** and **merges** from the **staging** to the **main branch**
- Issue: "**For Deploy**" → "**Done**"
- Code is deployed to the **production environment**
 - Often done manually



- **Issues flow** describes the **issue progress** over the time

Pending → Sprint TODO → In Progress → For Testing → Verified → For Deploy → Done

- **GitHub branches flow**

- **Feature** branch → **QA** branch (*qa*)
- **Feature** branch → **staging** branch (*staging*) → **production** branch (*main*)

- **Dev environments flow**

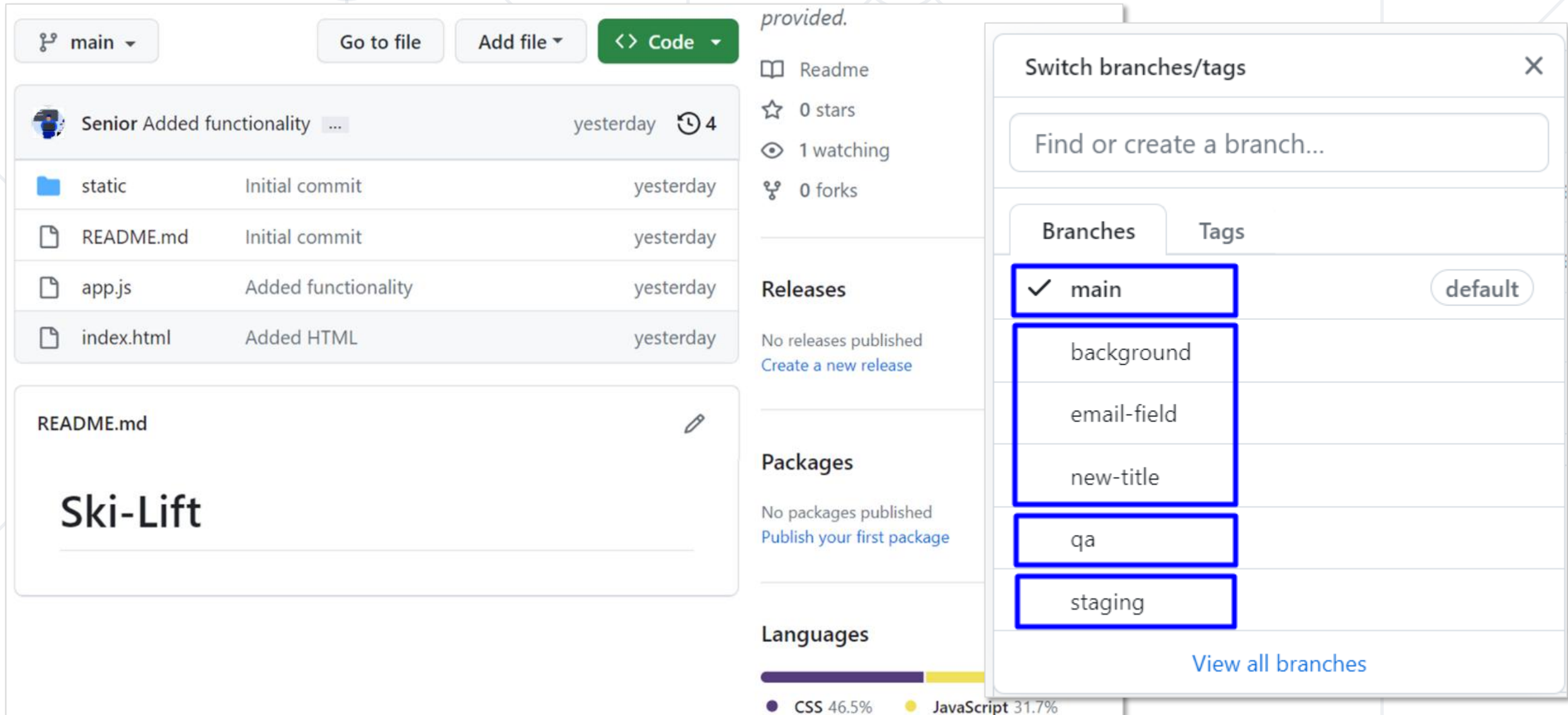
- Feature branch → **local** laptop (localhost)
- QA branch → **QA** environment (qa.myapp.com) – auto deploy by CI/CD
- Staging branch → **staging** environment (staging.myapp.com)
- Main branch → **production** environment (myapp.com) – manual deploy



Development Workflow in Action

Sample GitHub Project

Development Workflow in Action



The screenshot shows a GitHub repository interface for a project named "Ski-Lift". At the top, there's a navigation bar with a "main" branch selector, "Go to file", "Add file", and a "Code" button. Below this, a commit history table lists recent changes by a user named "Senior". To the right of the commit history, repository statistics are shown: 0 stars, 1 watching, and 0 forks. Further down, sections for "Releases" and "Packages" indicate no published content yet. At the bottom, a "Languages" section shows a bar chart with CSS at 46.5% and JavaScript at 31.7%.

On the right side, a "Switch branches/tags" modal is open, displaying a search bar and a list of branches. The "main" branch is selected and marked as "default". Other branches listed include "background", "email-field", "new-title", "qa", and "staging". A "View all branches" link is at the bottom of the modal.

File	Commit Message	Time
static	Initial commit	yesterday
README.md	Initial commit	yesterday
app.js	Added functionality	yesterday
index.html	Added HTML	yesterday

Releases
No releases published
[Create a new release](#)

Packages
No packages published
[Publish your first package](#)

Languages

Language	Percentage
CSS	46.5%
JavaScript	31.7%

Switch branches/tags

Find or create a branch...

Branches | **Tags**

- ✓ main (default)
- background
- email-field
- new-title
- qa
- staging

[View all branches](#)

Development Workflow in Action

Switch branches/tags

Find or create a branch...

Branches

Tags

- ✓ main
- background
- email-field
- new-title
- qa
- staging

[View all branches](#)

← ↻ 🏠 🔒 https://ski-lift.onrender.com

ski lift

← ↻ 🏠 🔒 https://qa-ski-lift.onrender.com

ski lift

↻ 🏠 🔒 https://staging-ski-lift.onrender.com

ski lift

***** BUY TICKETS *****

Buy ticket

First name:

Last name:

Number of people:

From Date:

Days:

Email

Next step ⇨

Ticket Preview

Confirm ticket

Development Workflow in Action

Ski Lift

View 1

+ New view

Filter by keyword or by field

Pending 3

This item hasn't been started

Ski-Lift #6

Room Type Field

new field

Ski-Lift #5

Price Field

duplicate new field

Ski-Lift #7

Room Type Card

bug duplicate

+ Add item

Sprint TODO 1

Ski-Lift #4

Added new color

+ Add item

In Progress 0

This is actively being worked on

+ Add item

For Testing 1

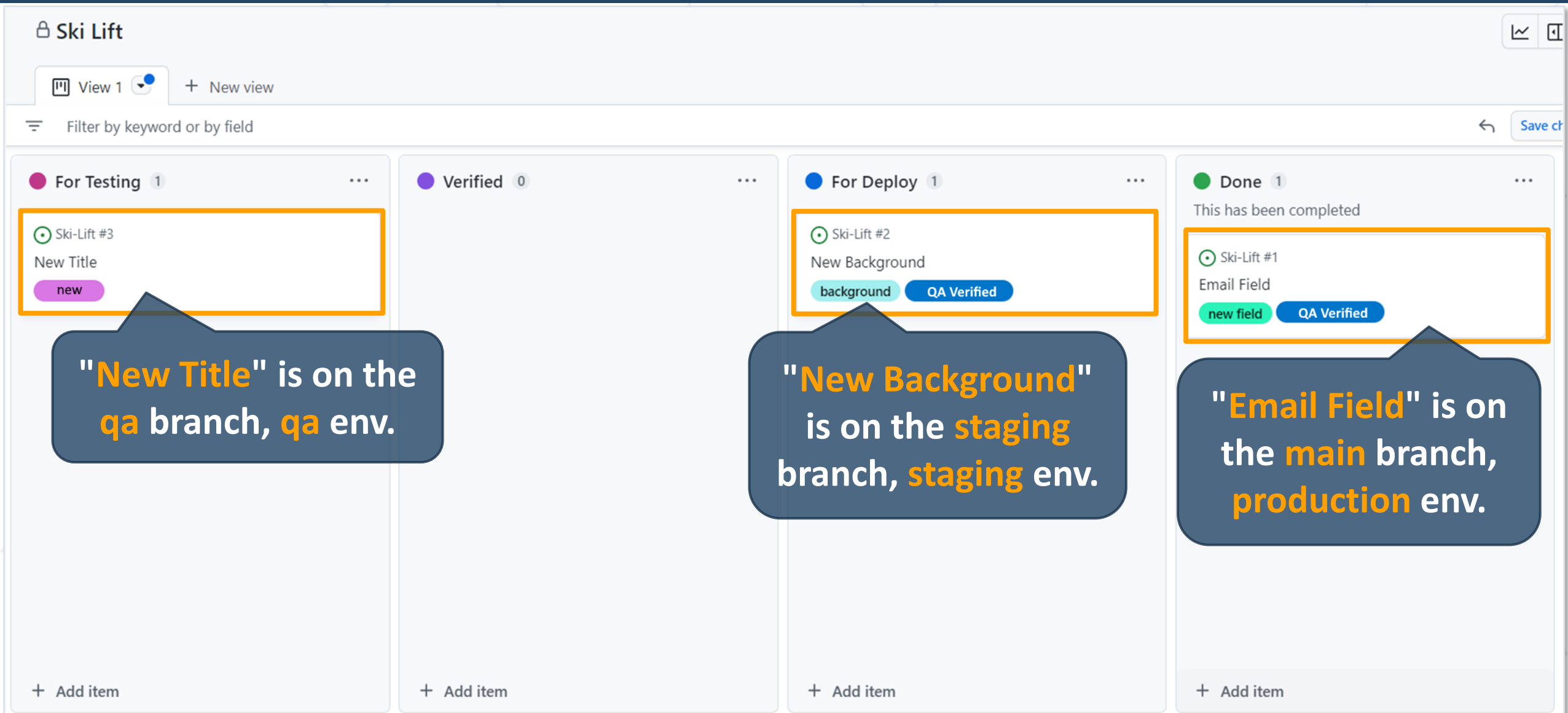
Ski-Lift #3

New Title

new

+ Add item

We have GitHub Projects Board with Issues



Ski Lift

View 1 + New view

Filter by keyword or by field

For Testing 1

- Ski-Lift #3
New Title
new

"New Title" is on the qa branch, qa env.

Verified 0

For Deploy 1

- Ski-Lift #2
New Background
background QA Verified

"New Background" is on the staging branch, staging env.

Done 1

This has been completed

- Ski-Lift #1
Email Field
new field QA Verified

"Email Field" is on the main branch, production env.

+ Add item

Email Field Issue

Email Field #1

Closed

Senior opened this issue 1 hour ago · 0 comments · Fixed by #2

Senior commented 1 hour ago · edited

Owner

...

TODO: add an **email field** in form.

As an end user, I want to be able to provide my email address when submitting a form so that I can **receive email notifications and updates related to my form submission**.

Overall, the addition of an email field to the form will enhance the user experience by providing them with a way to **receive important notifications** related to their form submission, and will also improve the functionality of the application by allowing for better **communication with users**.

Senior added the **new field** label 1 hour ago

Senior added this to **Ski Lift** 1 hour ago

Senior linked a pull request 59 minutes ago that will close this issue

Email field #2

Senior moved this from Pending to Sprint TODO in **Ski Lift** 58 minutes ago

Assignees

No one—assign yourself

Labels

new field **QA Verified**

Projects

Ski Lift

Status: Done

Milestone

No milestone

Development

Successfully merging a pull request may close this issue.

email-field

senior/Ski-Lift-Demo-Practice

Email field

senior/Ski-Lift-Demo-Practice

Merged

33

Email Field Issue

Closed

Email Field #1

Senior opened this issue 1 hours ago · 0 comments · Fixed by #2

Junior

added 2 commits 48 minutes ago

Added HTML

Added functionality

Junior

merged commit `4a149af` into `qa` 45 minutes ago

Junior

moved this from In Progress to For Testing in Ski Lift 43 minutes ago

Qa

added the QA Verified label 35 minutes ago

Qa

moved this from For Testing to Verified in Ski Lift 35 minutes ago

Revert

Unsubscribe

You're receiving notifications because you're watching this repository.

a1518b2

a3b658f

3 participants

Lock conversation

Pin issue ⓘ

→ Transfer issue

🗑 Delete issue

34

Email Field Issue

 Closed

Email Field #1

Senior opened this issue 1 hours ago · 0 comments · Fixed by #2

 Junior merged commit **6a189ag** into **staging** 30 minutes ago

 Revert

 Junior moved this from **Verified** to **For Deploy** in **Ski Lift** 30 minutes ago

 Senior merged commit **8c246af** into **main** 15 minutes ago

 Revert

 Senior moved this from **For Deploy** to **Done** in **Ski Lift** 12 minutes ago

 Senior closed this as completed 12 minutes ago

Email Field Issue

Closed

Email Field #1

Senior opened this issue 1 hours ago · 0 comments · Fixed by #2

Junior merged commit 6a189ag into staging 30 minutes ago

Junior moved this from Verified to For Deploy in Ski Lift 30 minutes ago

Senior merged commit 8c246af into main 15 minutes ago

Senior moved this from For Deploy to Done in Ski Lift 12 minutes ago

Senior closed this as completed 12 minutes ago

https://staging-ski-lift.onrender.com

Buy ticket

First name:

Last name:

Number of people:

From Date:

mm / dd / yyyy

Days:

Email

Next step ⇌

Email Field Issue

Closed

Email Field #1

Senior opened this issue 1 hours ago · 0 comments · Fixed by #2

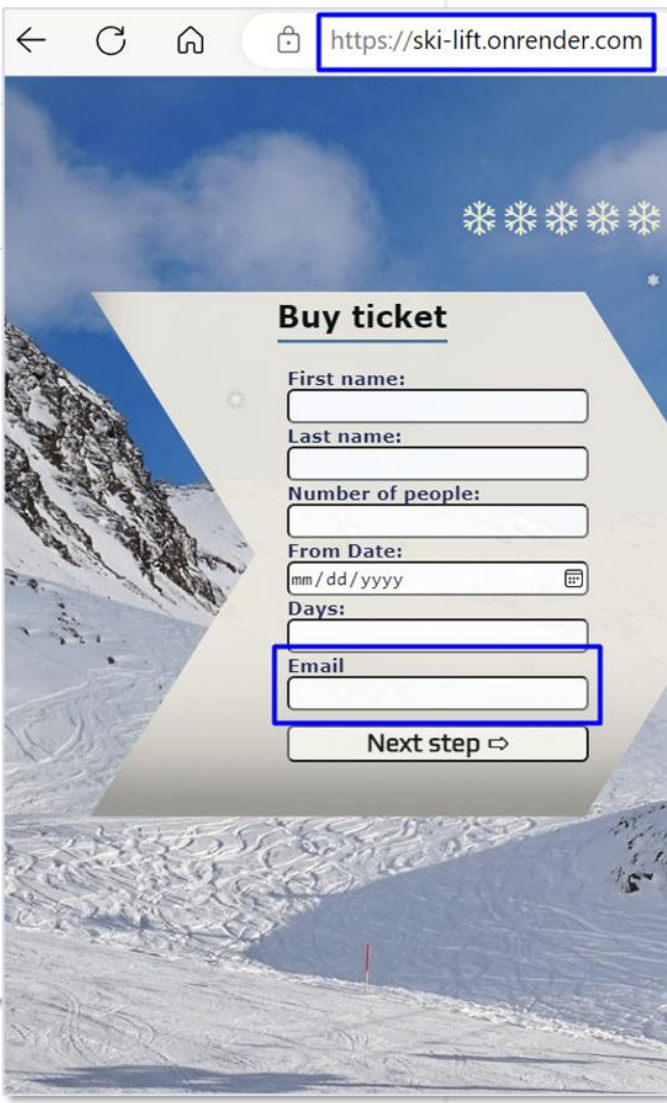
Junior merged commit 6a189ag into staging 30 minutes ago

Junior moved this from Verified to For Deploy in Ski Lift 30 minutes ago

Senior merged commit 8c246af into main 15 minutes ago

Senior moved this from For Deploy to Done in Ski Lift 12 minutes ago

Senior closed this as completed 12 minutes ago



New Background Issue

New Background #2

Open

Senior opened this issue 1 hours ago · 0 comments ·

Senior commented 1 hours ago · edited

Owner

...

TODO: Change the **background image**.

As an end user, I want to see a visually appealing background image on the website so that it creates a pleasant and engaging experience while using the website.

Overall, changing the background image will improve the aesthetics of the website, enhance the user experience, and make the website more engaging and visually appealing to users.

😊

🏷️ Senior added the **background** label 1 hours ago

📅 Senior added this to **Ski Lift** 1 hours ago

🔖 Senior linked a pull request 1 hour ago that will close this issue

Background image#3

🔗 Senior moved this from Pending to Sprint TODO in **Ski Lift** 1 hours ago

Assignees

No one—assign yourself

Labels

background **QA Verified**

Projects

📅 Ski Lift

Status: Done

Milestone

No milestone

Development

Successfully merging a pull request may close this issue.

background-image
senior/Ski-Lift-Demo-Practice

Background image
senior/Ski-Lift-Demo-Practice

🔗 Open

New Background Issue

Open

New Background #2
Senior opened this issue 1 hour ago · 0 comments

📄

Junior added 2 commits 36 minutes ago

🔗

Added img

Changes in code

🔗

Junior merged commit **9a149af** into **qa** 36 minutes ago

📄

Junior moved this from In Progress to For Testing in **Ski Lift** 34 minutes ago

🏷️

Qa added the **QA Verified** label 12 minutes ago

📄

Qa moved this from For Testing to Verified in **Ski Lift** 11 minutes ago

Notifications

Customize

🔔 Unsubscribe

a1518c2

a5b658f

You're receiving notifications because you're watching this repository.

Revert

3 participants

🔒 Lock conversation

📌 Pin issue ⓘ

➡ Transfer issue

🗑 Delete issue

39

New Background Issue

Open

New Background #2

Senior opened this issue 1 hour ago · 0 comments



Junior added 2 commits 36 minutes ago



Added img



Changes in code



Junior merged commit **9a149af** into **qa** 36 minutes ago



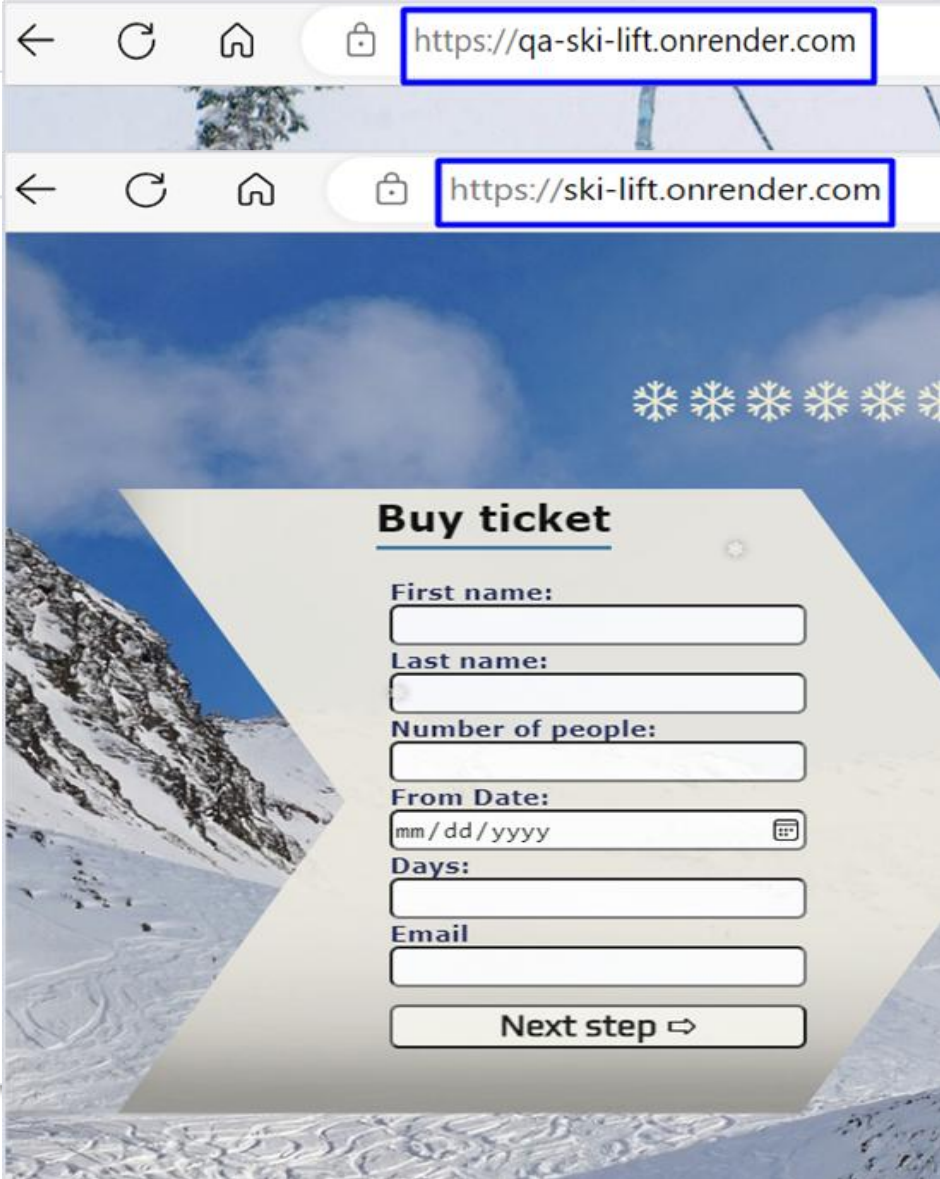
Junior moved this from In Progress to For Testing in Ski Lift 34 minutes ago



Qa added the **QA Verified** label 12 minutes ago



Qa moved this from For Testing to Verified in Ski Lift 11 minutes ago



← ↻ 🏠 🔒 <https://qa-ski-lift.onrender.com>

← ↻ 🏠 🔒 <https://ski-lift.onrender.com>

Buy ticket

First name:

Last name:

Number of people:

From Date: 📅

Days:

Email:

Next step ➡

New Title Issue

New Title #3

Open

Senior opened this issue 1 hours ago · 0 comments

Senior commented 1 hours ago · edited

Owner

...

TODO: Change the **color** of title.

As an end user, I want to be able to easily **read** and **distinguish** the **title** of the page from the rest of the content, so that I can quickly identify what the page is about.

Overall, changing the **color** of the title will **improve** the **aesthetics** and **readability** of the website, making it easier for users to identify the main purpose of the page and navigate the website more efficiently.

Senior added the **new**

abel 1 hours ago

Senior added this to **Ski Lift**

1 hours ago

Senior linked a pull request 1 hour ago that will close this issue

New title#4

Senior moved this from Pending to Sprint TODO in **Ski Lift**

1 hours ago

Assignees

No one—assign yourself

Labels

new

Projects

Ski Lift

Status: Done

Milestone

No milestone

Development

Successfully merging a pull request may close this issue.

new-title

senior/Ski-Lift-Demo-Practice

New Title

senior/Ski-Lift-Demo-Practice

Open

New Title Issue

Open

New Title #3
Senior opened this issue 1 hour ago · 0 comments

📄

Junior added 2 commits 36 minutes ago

🔗

Changes in HTML

Changes in code

🔄

Junior merged commit **9a149af** into **qa** 35 minutes ago

📁

Junior moved this from **In Progress** to **For Testing** in **Ski Lift** 34 minutes ago

Revert

Notifications

Customize

🔔

Unsubscribe

a1518c2

a5b658f

You're receiving notifications because you're watching this repository.

3 participants

🔒

Lock conversation

42

- **Issues** maintain an **organized** software development **workflow**
 - Moved across **board columns** to **visualize progress**
- Branches and environments **feature** (**local** env.), **qa** (**QA** env.), **staging** (**staging** env), **main** (**production** env.)
- Software development workflow **steps** (sprint planning, assign to developer, assign to QA, QA review, code review, deploy to production)



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